

-- HDOC DAY – 6 : ACTIVITY REPORT --

QUESTION 1 ; apparent weight calculation.

1. Problem statement identification

- Problem statement – write a C program to calculate the apparent weight of an object submerged in a fluid when its actual weight and the upthrust acting on it are provided.

- Input :

- actual weight (w)

- Upthrust (U)

- Output :

- Apparent weight (W app)

- Formula :

- Apparent weight = actual weight – upthrust

- $W_{app} = W - U$

2. TEST CASE TABLE

Test case	Input(s)	Expected output(s)	Actual output(s)	Result (pass/fail)
1.	Actual weight = 50, Upthrust = 10	40.000000	40.000000	pass
2.	Actual weight = 100.5, Upthrust = 25.5	75.000000	75.000000	pass
3.	Actual weight = 15,	0.000000	0.000000	pass

	Upthrust = 15			
--	------------------	--	--	--

3. Algorithm

- Declare variables ; *actualweight* , *upthrust* , and *apparent_weight*.
- Ask the user to enter the actual weight and upthrust values.
- Read the values of *actual_weight* and *upthrust*.
- Calculate *apparent_weight* = *actual_weight* – *upthrust*
- Display the calculated *apparent_weight* value up to two decimal places.
- Stop the code.

4. C program :

```
#include <stdio.h>
```

```
int main() {
```

```
    float actual, upthrust, apparent;
```

```
    printf("Enter actual weight: ");
```

```
    scanf("%f", &actual);
```

```
    printf("Enter upthrust: ");
```

```
    scanf("%f", &upthrust);
```

```
    apparent = actual - upthrust;
```

```
printf("Apparent weight is: %f\n", apparent);

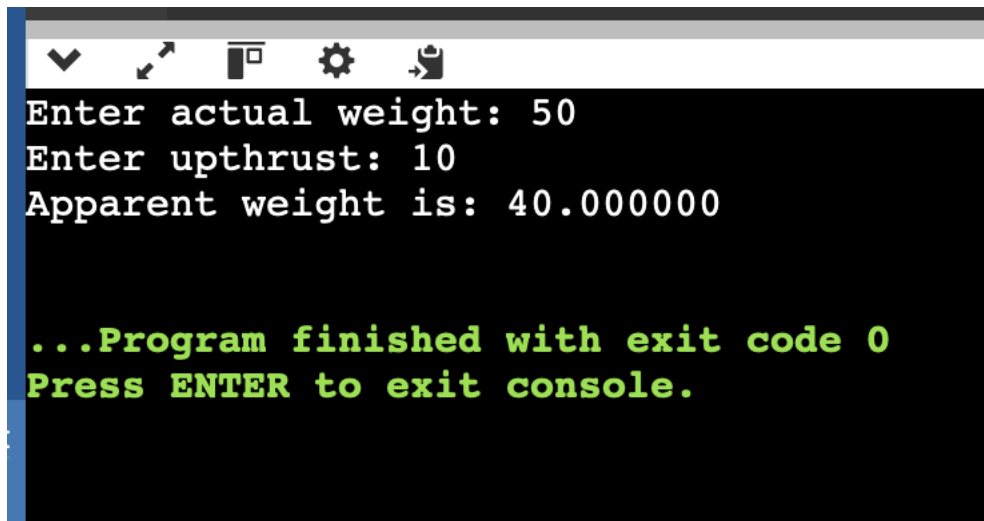
return 0;

}
```

5. Code execution :

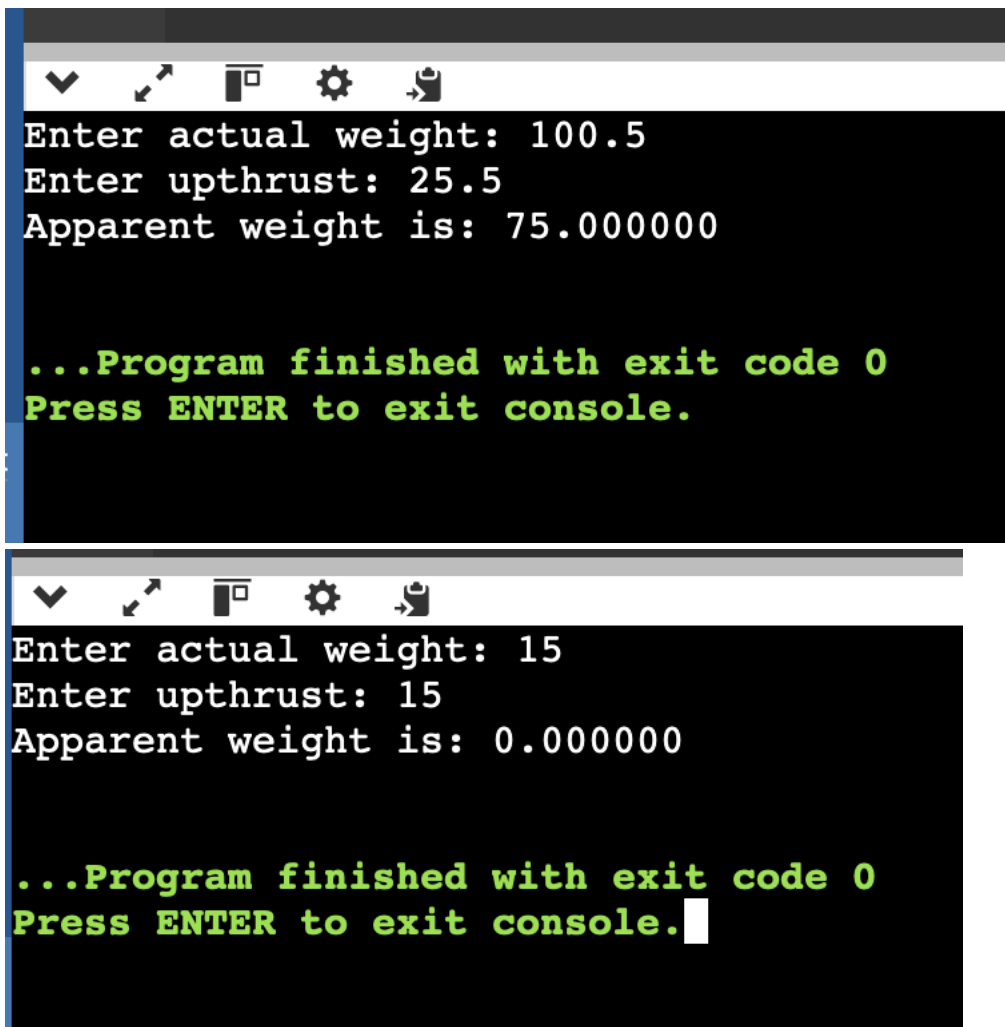
- The program executed successfully across all three test cases, giving outputs identical to the expected mathematical values.

6. Screenshot :

A screenshot of a console window with a black background and white text. The window has a title bar with standard icons (minimize, maximize, close, settings, and a taskbar icon). The text in the console reads: "Enter actual weight: 50", "Enter upthrust: 10", and "Apparent weight is: 40.000000". At the bottom, there is a green text message: "...Program finished with exit code 0" and "Press ENTER to exit console.".

```
Enter actual weight: 50
Enter upthrust: 10
Apparent weight is: 40.000000

...Program finished with exit code 0
Press ENTER to exit console.
```



```
Enter actual weight: 100.5
Enter upthrust: 25.5
Apparent weight is: 75.000000

...Program finished with exit code 0
Press ENTER to exit console.

Enter actual weight: 15
Enter upthrust: 15
Apparent weight is: 0.000000

...Program finished with exit code 0
Press ENTER to exit console.
```

QUESTION 2 : Latent heat calculation

1. Problem statement identification

- Problem statement ; write a C program to calculate the specific latent heat of the substance when total heat energy supplied and the mass of the substance are given.
- Input :
 - Heat energy (Q)
 - Mass (m)

- Output ;
 - Latent heat (L)
- Formula :
 - latent heat = Heat energy/mass
 - $L = Q/m$

2. TEST CASE TABLE

Test case	Input(s)	Expected output(s)	Actual output	Result (pass/fail)
1.	Heat energy = 500, Mass = 5	100.000000	100.000000	pass
2.	Heat energy = 2260, Mass = 2	1130.000000	1130.000000	pass
3.	Heat energy = 334, Mass = 1	334.000000	334.000000	pass

3. ALGORITHM ;

- Start the code.
- Declare variables : *heat_energy, mass and latent_heat*
- Read the inputs given by user for *heat_energy* and *mass*
- Check if the given mass by the user is greater than zero to avoid division zero error.
- Calculate *latent_heat = heat_energy/mass*
- *Print the calculated latent_heat*
- *Stop the code.*

4. C program :

```
#include <stdio.h>

int main() {
    float heat, mass, latent;

    printf("Enter heat energy: ");
    scanf("%f", &heat);

    printf("Enter mass: ");
    scanf("%f", &mass);

    latent = heat / mass;

    printf("Latent heat is: %f\n", latent);

    return 0;
}
```

5. Code Execution :

- The program compiled successfully without warnings.
- All test cases were calculated , and exact outputs were obtained by the mathematical formula.

6. Screenshots :



```
Enter heat energy: 500
Enter mass: 5
Latent heat is: 100.000000
```

```
...Program finished with exit code 0
Press ENTER to exit console.
```



```
Enter heat energy: 2260
Enter mass: 2
Latent heat is: 1130.000000
```

```
...Program finished with exit code 0
Press ENTER to exit console.
```



```
Enter heat energy: 334
Enter mass: 1
Latent heat is: 334.000000
```

```
...Program finished with exit code 0
Press ENTER to exit console.
```

QUESTION 3 ; Energy stored in a capacitor

1. Problem Statement Identification

- Problem statement – write a C program to calculate the energy stored in a capacitor when its capacitance and potential difference (voltage) are given.

- Input :
- Capacitance (C)
- Voltage (V)

- Output ;
- Stored energy (E)

- Formula ;
 - $\text{Energy} = 0.5 * \text{Capacitance} * \text{Voltage}^2$
 - $E = 0.5 * C * V^2$

2. Test case table ;

Test case	Input(s)	Expected output(s)	Actual output(s)	Result (pass/fail)
1.	Capacitance = 2, Voltage = 5	25.000000	25.000000	pass
2.	Capacitance = 4, Voltage = 10	200.000000	200.000000	pass
3.	Capacitance = 0.5, Voltage = 6	9.000000	9.000000	pass

3. ALGORITHM :

- Start.
- Declare variables : capacitance , voltage and energy.
- Enter the values of inputs of capacitance and voltage.
- Calculate the stored energy using the equation : $\text{energy} = 0.5 * \text{capacitance} * \text{voltage} * \text{voltage}$
- Display the output.
- Stop the code.

4. C program

```
#include <stdio.h>

int main() { float c, v, energy;

printf("Enter capacitance: ");
scanf("%f", &c);

printf("Enter voltage: ");
scanf("%f", &v);

energy = 0.5 * c * v * v;

printf("Energy stored is: %f\n", energy);

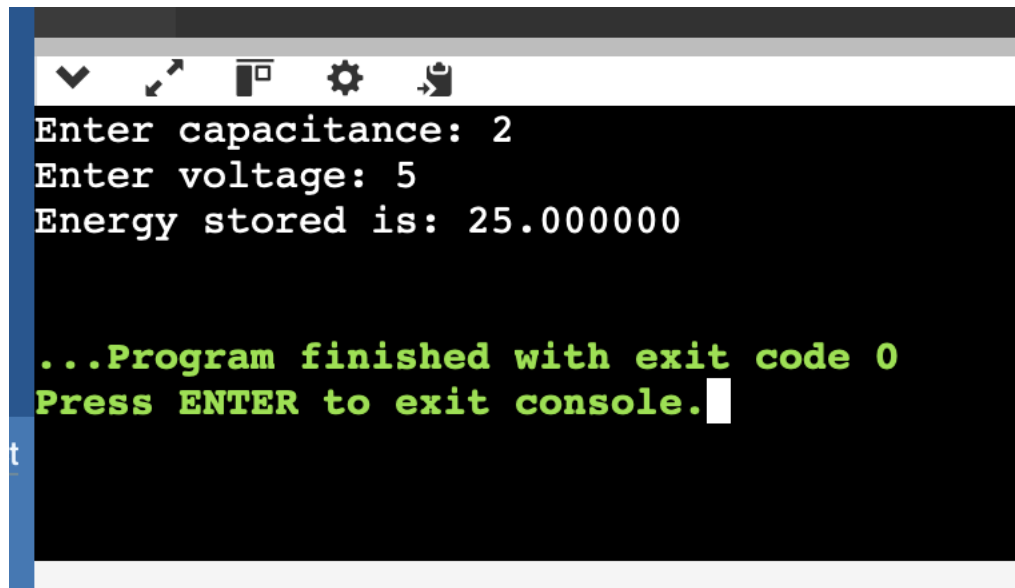
return 0;

}
```

5. Code execution :

- The code compiled successfully .
- Validate with test inputs ; the computed values matched expected output parameters perfectly.

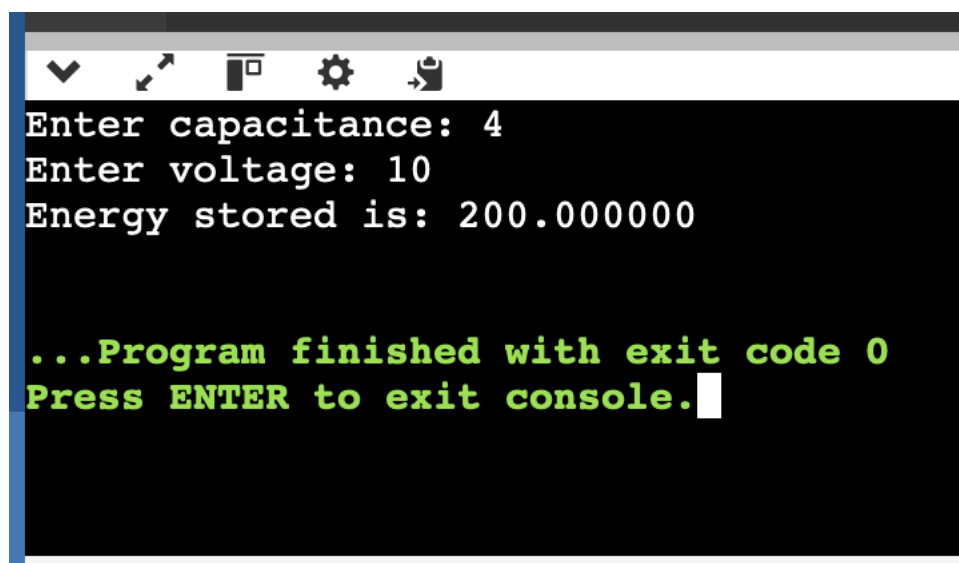
6. Screenshots :



A screenshot of a terminal window with a dark background and a light gray title bar. The title bar contains several icons: a downward arrow, a magnifying glass, a square, a gear, and a document. The terminal text is as follows:

```
Enter capacitance: 2
Enter voltage: 5
Energy stored is: 25.000000

...Program finished with exit code 0
Press ENTER to exit console.
```



A screenshot of a terminal window with a dark background and a light gray title bar. The title bar contains several icons: a downward arrow, a magnifying glass, a square, a gear, and a document. The terminal text is as follows:

```
Enter capacitance: 4
Enter voltage: 10
Energy stored is: 200.000000

...Program finished with exit code 0
Press ENTER to exit console.
```



```
Enter capacitance: 0.5  
Enter voltage: 6  
Energy stored is: 9.000000
```

```
...Program finished with exit code 0  
Press ENTER to exit console.
```